BLE-MIRABILIS-BLUE

Reader-friendly GATT Table & Protocol Reference - Firmware v0.6.2

Scope: all implemented GATT functionality except File Count (...000D). Total Uploaded Bytes (...000E) is included.

1. Standard Device Information Service

Service UUID: **0x180A**

Characteristic	UUID	Properties	Security	Value / behavior
Serial Number String	0x2A25	READ	Open	Generated from the nRF52840 hardware device ID; uppercase hexadecimal string.
Hardware Revision String	0x2A27	READ	Open	`NICE-NANO-COMPATIBLE`
Firmware Revision String	0x2A26	READ	Open	`0.6.2`

2. Custom Tutorial Service

Service UUID: **7E57A000-0000-4B1A-9C00-000000000001**

Name	UUID	Properties	Security	Purpose / behavior
Basic Write	...0002	WRITE	Open	Accepts bytes; each write replaces the previously stored value.
Last Written Value	...0003	READ	Open	Returns latest ...0002 value; before any write returns ASCII **N/A**.
Observable Write	...0004	WRITE	Open	Changes the state exposed by ...0005.
Observable Value	...0005	READ | NOTIFY	Open	READ returns current ...0004 value (or N/A); subscribers are notified after ...0004 changes it.
Periodic Event Stream	...0006	NOTIFY	Open	Notify-only: one-byte sequential counter every 2 seconds after subscription.
Write Without Response	...0007	WRITE NO RSP	Open	Accepts bytes using ATT Write Command, with no ATT write response.
Last Write-No-Response Value	...0008	READ	Open	Returns latest ...0007 value; before any write returns ASCII N/A.

3. Pairing / Encrypted Characteristics

These attributes require an encrypted BLE connection. Access from an unencrypted link can cause the central to establish security/pairing.

Name	UUID	Properties	Required access	Behavior
Secure Write	...0009	WRITE	Encrypted write	Stores supplied bytes; successful changes are reflected by ...000A.
Secure State	...000A	READ | NOTIFY	Encrypted read + CCCD	READ returns secure value (or N/A); subscribers receive updates after ...0009 writes.
File Transfer RX	...000B	WRITE NO RSP	Encrypted write	Central -> peripheral: upload requests/chunks and download ACK/NACK/CAN.
File Transfer TX	...000C	NOTIFY	Encrypted CCCD	Peripheral -> central: download chunks and upload ACK/NACK.
Total Uploaded Bytes	...000E	READ	Encrypted read	64-bit little-endian cumulative successful-upload payload bytes since boot.

4. File Transfer Protocol

Encrypted Write Without Response on RX plus encrypted notification subscription on TX.

Direction	Characteristic	GATT operation	Carries
Central -> Peripheral	...000B FILE_TRANSFER_RX	WRITE WITHOUT RESPONSE	Upload request/chunks; download ACK/NACK/CAN.
Peripheral -> Central	...000C FILE_TRANSFER_TX	NOTIFY	Download chunks; upload ACK/NACK.

Token	Meaning
0x02 STR	Normal data chunk.
0x03 ETX	Final data chunk.
0x06 ACK	Batch accepted.
0x15 NACK	Batch/chunk problem; retransmission path.
0x18 CAN	Cancel / abort.
'U'	Start upload; optional uint32 little-endian expected size follows.
'D'	Start download of the currently stored RAM-backed file.

Chunk format

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Byte 0 = STR/ETX | Byte 1 = sequence | Bytes 2..19 = payload
```

20 bytes total: 2 protocol bytes + up to 18 payload bytes. Sequence is uint8 and rolls over after 0xFF. Up to 8 chunks are grouped per acknowledgement.

5. Storage Behavior & Reader's Test Map

One file is stored in a 16 KiB RAM buffer. A new successful upload replaces it. Disconnecting does not delete it; reset/power loss does. Downloading does not remove it. TOTAL_UPLOADED_BYTES is cumulative since boot.

Concept	Characteristics	Suggested test
Standard READ	2A25 / 2A27 / 2A26	Read identity and revision strings.
WRITE -> READ	...0002 -> ...0003	Write HEX bytes, then read them back.
WRITE -> READ + NOTIFY	...0004 -> ...0005	Subscribe to ...0005; write ...0004; observe notify and read-back.
Notify-only	...0006	Subscribe and observe sequential events every 2 seconds.
WRITE NO RSP -> READ	...0007 -> ...0008	Send bytes without ATT response, then read stored value.
Pairing / encryption	...0009 / ...000A	Access before pairing, establish security, then test secure write/read/notify.
File upload	...000B + ...000C	Subscribe TX; send U + chunks on RX; observe ACKs.
File download	...000B + ...000C	Send D on RX; receive chunks on TX; ACK each batch on RX.
Transfer metric	...000E	Read cumulative uploaded bytes as uint64 little-endian.

UUID shorthand

...0002 through ...000E abbreviate `7E57A000-0000-4B1A-9C00-0000000000xx`. File Count (...000D) is intentionally excluded.